



CAMBRIDGE
International Education



Three Rivers Academy

Cambridge International AS & A Level

CHEMISTRY

9701/31

Paper 3 Advanced Practical Skills 1

June 2026

CONFIDENTIAL INSTRUCTIONS

This document gives details of how to prepare for and administer the practical exam.

The information in this document and the identity of any materials supplied by Cambridge International are confidential and must NOT reach candidates either directly or indirectly.

The supervisor must complete the report at the end of this document and return it with the scripts.

INSTRUCTIONS

- If you have any queries regarding these confidential instructions, contact Cambridge International stating the centre number, the syllabus and component number and the nature of the query.
email info@cambridgeinternational.org
phone +44 1223 553554

This document has **8** pages. Any blank pages are indicated.

General information about practical exams

Centres must follow the guidance on science practical exams given in the *Cambridge Handbook*.

Safety

Supervisors must follow national and local regulations relating to safety and first aid.

Only those procedures described in the question paper should be attempted.

Supervisors must inform candidates that materials and apparatus used in the exam should be treated with caution. Suitable eye protection should be used where necessary.

The following hazard codes are used in these confidential instructions, where relevant:

C	corrosive	MH	moderate hazard
HH	health hazard	T	acutely toxic
F	flammable	O	oxidising
N	hazardous to the aquatic environment		

Hazard data sheets relating to substances used in this exam should be available from your chemical supplier.

Before the exam

- The packets containing the question papers must **not** be opened before the exam.
- It is assumed that standard school laboratory facilities, as indicated in the *Guide to Planning Practical Science*, will be available.
- Spare materials and apparatus for the tasks set must be available for candidates, if required.

During the exam

- It must be made clear to candidates at the start of the exam that they may request spare materials and apparatus for the tasks set.
- Where specified, the supervisor **must** perform the experiments and record the results as instructed. This must be done **out of sight** of the candidates, using the same materials and apparatus as the candidates.
- Any assistance provided to candidates must be recorded in the supervisor's report.
- If any materials or apparatus need to be replaced, for example, in the event of breakage or loss, this must be recorded in the supervisor's report.

After the exam

- The supervisor must complete a report for each practical session held and each laboratory used.
- Each packet of scripts returned to Cambridge International must contain the following items:
 - the scripts of the candidates specified on the bar code label provided
 - the supervisor's results relevant to these candidates
 - the supervisor's reports relevant to these candidates
 - seating plans for each practical session, referring to each candidate by candidate number
 - the attendance register.

Specific information for this practical exam

During the exam, the supervisor (**not** the invigilator) must do all the experiments and record the results on a spare copy of the question paper, clearly labelled 'supervisor's results'.

If chemicals are prepared in more than one batch, clearly labelled supervisor's results must be provided for each batch. The candidates using each batch must be listed on the supervisor's report.

Apparatus

The apparatus listed must be provided to each candidate.

1 × 25 cm³ pipette
 1 × pipette filler
 1 × 50 cm³ burette
 2 × 150 cm³ or 250 cm³ conical flask
 1 × 50 cm³ measuring cylinder
 1 × burette stand and clamp
 1 × 100 cm³ beaker
 1 × 250 cm³ beaker
 1 × funnel (for filling burette)
 1 × white tile
 1 × thermometer (−10 °C to +110 °C at 1 °C)
 2 × plastic or cardboard cup, capacity approximately 150 cm³
 1 × glass rod
 2 × teat/dropping pipette
 1 × spatula
 1 × Bunsen burner
 1 × heat-proof mat
 1 × test-tube holder
 1 × boiling tube
 8 × test-tube*
 1 × hard-glass test-tube
 1 × test-tube rack
 balance, single-pan, direct reading, minimum accuracy 0.01 g (1 per 8–12 candidates) weighing to 200 g
 1 × wash bottle containing distilled water
 1 × pen for labelling glassware
 paper towels
 red and blue litmus papers
 aluminium foil
 wooden splints
 the apparatus normally used in the centre for use with limewater in testing for carbon dioxide

*Candidates are expected to rinse and reuse test-tubes where possible.
 Additional test-tubes should be available.

Materials

The materials listed in the table must be provided to each candidate. **Materials must be labelled only as specified in the 'label' column. The identities of chemicals labelled with letter codes, e.g. FA 1, may be different from their descriptions in the question paper. Candidates must use the descriptions given in the question paper. For example, candidates may be supplied with sulfuric acid, labelled as FA 1, but be told in the question paper that FA 1 is phosphoric acid.**

label	per candidate	identity	notes
FA 1 [MH]	7.5 g	hydrated sodium carbonate	Supply 7.4–7.6 g of $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ [MH] in a stoppered container. A freshly-opened bottle of $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ should be used.
FA 2 [MH]	80 cm ³	3.00 mol dm ⁻³ hydrochloric acid	Dilute 255 cm ³ of concentrated HCl (35% w/w) [C][MH] with distilled water to 1.00 dm ³
FA 3 [MH]	3.1 g	anhydrous sodium carbonate	Supply 3.0–3.2 g of Na_2CO_3 [MH] in a stoppered container. A freshly-opened bottle of Na_2CO_3 should be used.
FA 4	120 cm ³	0.0519 mol dm ⁻³ sodium carbonate	Dissolve 5.50 g of Na_2CO_3 [MH] in each dm ³ of distilled water. Supply in a stoppered bottle. (The description on the question paper is different.)
FA 5	120 cm ³	0.110 mol dm ⁻³ hydrochloric acid	Dilute 36.7 cm ³ of FA 2 [MH] with distilled water to 1.00 dm ³ . Supply in a stoppered bottle.
FA 6 [F][MH][HH]	10 cm ³	bromophenol blue indicator	Prepare as described in the current syllabus. Supply in a dropping bottle.
FA 7 [MH]	1.0 g	ammonium carbonate (solid)	Supply 0.9–1.1 g of $(\text{NH}_4)_2\text{CO}_3$ [MH] in a stoppered container
FA 8	25 cm ³	0.25 mol dm ⁻³ ammonium carbonate	Dissolve 24.0 g of $(\text{NH}_4)_2\text{CO}_3$ [MH] in each dm ³ of distilled water. Some suppliers may give a different formula for this material.
FA 9 [C]	25 cm ³	2.0 mol dm ⁻³ sodium hydroxide	Dissolve 80.0 g of NaOH [C] in each dm ³ of distilled water. Supply in a stoppered bottle.
aqueous zinc sulfate [MH][N]	10 cm ³	0.10 mol dm ⁻³ zinc sulfate	Dissolve 28.8 g of $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ [C][MH][N] in each dm ³ of distilled water.

label	per candidate	identity	notes
dilute hydrochloric acid	10 cm ³	2.0 mol dm ⁻³ HCl	See preparation instructions in the current syllabus. If necessary, each of these reagents can be provided as a communal supply for groups of up to 6 candidates. Invigilators must be alert to the risk of contamination and the opportunity for malpractice when using a communal supply.
dilute nitric acid [C]	10 cm ³	2.0 mol dm ⁻³ HNO ₃	
dilute sulfuric acid [MH]	10 cm ³	1.0 mol dm ⁻³ H ₂ SO ₄	
aqueous ammonia [C][MH][N]	10 cm ³	2.0 mol dm ⁻³ NH ₃	
aqueous sodium hydroxide [C]	10 cm ³	1.0 mol dm ⁻³ NaOH	
aqueous barium chloride or aqueous barium nitrate	10 cm ³	0.1 mol dm ⁻³ BaCl ₂ or 0.1 mol dm ⁻³ Ba(NO ₃) ₂	
limewater [MH]	10 cm ³	saturated aqueous calcium hydroxide, Ca(OH) ₂	
aqueous silver nitrate	10 cm ³	0.05 mol dm ⁻³ AgNO ₃	
acidified aqueous potassium manganate(VII) [MH]	10 cm ³	0.01 mol dm ⁻³ KMnO ₄ in 0.5 mol dm ⁻³ H ₂ SO ₄	

- An excess of at least 10% of each material must be prepared to cover accidental loss.
- All solutions must be thoroughly mixed.
- If you are unable to source any of these chemicals, you must contact Cambridge International as far as possible in advance of the exam for advice.